

Lam M. Nguyen

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EDUCATION

- 2014 – 2018 **Ph.D.**, Department of Industrial and Systems Engineering, *Lehigh University*, Bethlehem, PA
Thesis advisors: *Katya Scheinberg*, *Martin Takac*, and *Alexander L. Stolyar*
Thesis title: A Service System with On-Demand Agents, Stochastic Gradient Algorithms and the SARAH Algorithm
Elizabeth V. Stout Dissertation Award
Research areas: Optimization for Large Scale Problems, Machine Learning, Deep Learning, Stochastic Models, Optimal Control
- 2011 – 2013 **M.B.A.**, College of Business, *McNeese State University*, Lake Charles, LA
Beta Gamma Sigma (Academic Honor)
- 2004 – 2008 **Bachelor**, Applied Mathematics and Computer Science, Faculty of Computational Mathematics and Cybernetics, *Lomonosov Moscow State University*, Moscow, Russia
Thesis advisor: *Vladimir I. Dmitriev*
Thesis title: Methods for Detecting Hidden Period in Some Economics Processes

RESEARCH EXPERIENCE

- 06/2022 – Present **Staff Research Scientist**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Reinforcement Learning, Time Series
- 04/2021 – 06/2022 **Research Staff Member**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Reinforcement Learning
- 09/2020 – 12/2025 **Principal Investigator**, *MIT-IBM Watson AI Lab*, Cambridge, MA
Research areas: Dynamical Systems, Time Series, Reinforcement Learning, Adversarial Robustness
- 10/2018 – 03/2021 **Research Scientist**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Deep Learning, Reinforcement Learning, AI Solutions, Explainable AI
- 05/2018 – 08/2018 **Research Intern**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Deep Learning, Reinforcement Learning
- 08/2017 – 05/2018 **Research Co-op**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Deep Learning
- 06/2017 – 08/2017 **Research Intern**, *IBM Research, Thomas J. Watson Research Center*, Yorktown Heights, NY
Research areas: Optimization, Machine Learning, Deep Learning
- 09/2014 – 05/2017 **Research Assistant**, *Lehigh University*, Bethlehem, PA
Research areas: Optimization for Large Scale Problems, Machine Learning, Deep Learning, Stochastic Models, Optimal Control
- 01/2012 – 12/2013 **Graduate (Research) Assistant**, *McNeese State University*, Lake Charles, LA
Research areas: Operations Management and Finance

EDITORSHIP / PROGRAM COMMITTEE / ORGANIZING COMMITTEE

CONFERENCE ORGANIZING COMMITTEE

2026	Lead Journal Chair , The 40th Conference on Neural Information Processing Systems (NeurIPS 2026)
2025	(Lead) Journal Chair , The 39th Conference on Neural Information Processing Systems (NeurIPS 2025)
2024	(Lead) Journal Chair , The 38th Conference on Neural Information Processing Systems (NeurIPS 2024)
2023	(Lead) Journal Chair , The 37th Conference on Neural Information Processing Systems (NeurIPS 2023)

EDITORSHIP (PEER-REVIEWED JOURNALS)

06/2022 – Present	Action Editor , Journal of Machine Learning Research
06/2021 – Present	Action Editor , Machine Learning
06/2022 – Present	Associate Editor , Journal of Optimization Theory and Applications
01/2022 – 12/2023	Associate Editor , IEEE Transactions on Neural Networks and Learning Systems
01/2022 – 12/2022	Action Editor , Neural Networks

SENIOR AREA CHAIR (PEER-REVIEWED CONFERENCES)

2024 – 2026	Senior Area Chair , Conference on Neural Information Processing Systems (NeurIPS)
2026	Senior Area Chair , International Conference on Machine Learning (ICML)
2026 – 2027	Senior Area Chair , International Conference on Learning Representations (ICLR)
2025 – 2027	Senior Area Chair , International Conference on Artificial Intelligence and Statistics (AISTATS)
2027	(Senior) Area Chair , AAAI Conference on Artificial Intelligence (AAAI)

AREA CHAIR / META-REVIEWER / SENIOR PROGRAM COMMITTEE (PEER-REVIEWED CONFERENCES)

2020 – 2025	Area Chair , International Conference on Machine Learning (ICML)
2022 – 2023	Area Chair , Conference on Neural Information Processing Systems (NeurIPS)
2021 – 2025	Area Chair , International Conference on Learning Representations (ICLR)
2021 – 2024	Area Chair , International Conference on Artificial Intelligence and Statistics (AISTATS)
2022 – 2024	Area Chair , Conference on Uncertainty in Artificial Intelligence (UAI)
2023 – 2025	Area Chair , Conference on Computer Vision and Pattern Recognition (CVPR)
2022	Senior Program Committee , AAAI Conference on Artificial Intelligence (AAAI)

WORKSHOP ORGANIZING COMMITTEE

2026	Program Chair , Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety across Modalities, ICLR 2026 Workshop
2023	General Chair & Program Chair , When Machine Learning meets Dynamical Systems: Theory and Applications, AAAI 2023 Workshop
2021	General Chair & Program Chair , New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership (NFFL 2021), NeurIPS 2021 Workshop

REVIEWER (PROPOSALS)

2022	Reviewer & Panelist , Grant proposals, National Science Foundation (NSF)
2022, 2024	Evaluation Member , Grant proposals, AI Singapore (AISG) Research Programme
2021	Reviewer , Workshop proposals, NeurIPS 2021 Workshops

GRANT EXPERIENCE

- 01/2026 – 12/2026 **Principal Investigator**, “*Time Series Data Agent: an Agentic System with Foundation Models for Multimodal Data*”, NAIRR Deep Partnership program “IBM & the AI Alliance - Collaboration on Open Source Cloud Platform, Models and Tools”, \$110K
IBM PI: **Lam M. Nguyen**, IBM Co-PI: Chandra Reddy
RPI PI: Agung Julius
RPI Students: Yunshi Wen, Nicolas Lizarralde
- 01/2025 – 12/2025 **Principal Investigator**, “*Interpretable Foundation Models for General-Purpose Time Series Analysis*”, RPI - IBM Future of Computing Research Collaboration, \$150K
IBM PI: **Lam M. Nguyen**
RPI PI: Agung Julius
RPI Students: Yunshi Wen, Nicolas Lizarralde
- 01/2023 – 12/2025 **Principal Investigator**, “*Safe Learning for Time Series Problems: Data, Structure and Optimization*”, MIT-IBM Watson AI Lab Foundational Project, \$750K + \$150K
IBM PI: **Lam M. Nguyen**, IBM Co-PI: Subhro Das
MIT PI: Luca Daniel, MIT Co-PI: Alexandre Megretski
MIT Students: Wang Zhang, Ziwen (Martin) Ma (Harvard)
- 01/2022 – 12/2022 **Principal Investigator**, “*Safe AI Certification*”, MIT-IBM Watson AI Lab Project, \$150K
IBM PI: **Lam M. Nguyen**, IBM Co-PI: Subhro Das
MIT PI: Alexandre Megretski, MIT Co-PI: Luca Daniel
MIT Student: Wang Zhang
- 01/2021 – 12/2021 **Principal Investigator**, “*Safety Structures, Certification, and Training for AI in the Feedback Loop*”, MIT-IBM Watson AI Lab Exploratory Project, \$150K.
IBM PI: **Lam M. Nguyen**, IBM Co-PI: Subhro Das, Tsui-Wei Weng
MIT PI: Alexandre Megretski, MIT Co-PI: Luca Daniel
MIT Student: Wang Zhang
- 09/2020 – 09/2021 **Co-Principal Investigator**, “*Hierarchical Disentangled Representations for Scalable Multi-agent Reinforcement Learning*”, MIT-IBM Watson AI Lab Exploratory Project, \$100K.
IBM PI: Tsui-Wei Weng, IBM Co-PI: **Lam M. Nguyen**
MIT PI: Cathy Wu
MIT Student: Vindula Jayawardana

BOOK

- [1] Federated Learning: Theory and Practice.
Lam M. Nguyen, Trong Nghia Hoang, Pin-Yu Chen
Elsevier publisher, ISBN 9780443190377, 2024

JOURNAL & PEER-REVIEWED CONFERENCE PAPERS

- [57] Revisiting the Generic Transformer: Deconstructing a Strong Baseline for Time Series Foundation Models.
Yunshi Wen, Wesley M. Gifford, Chandra Reddy, **Lam M. Nguyen**, Jayant Kalagnanam, Anak Agung Julius
The 40th Conference on Neural Information Processing Systems (NeurIPS 2026), 2026
- [56] Adjusted Shuffling SARAH: Advancing Complexity Analysis via Dynamic Gradient Weighting.
Duc Toan Nguyen, Trang H. Tran, **Lam M. Nguyen**
Journal of Optimization Theory and Applications (JOTA), 2026
- [55] Graph Concept Bottleneck Models.
Haotian Xu, Tsui-Wei Weng, **Lam M. Nguyen**, Tengfei Ma
Transactions on Machine Learning Research (TMLR), 2026

- [54] Stochastic ISTA/FISTA Adaptive Step Search Algorithms for Convex Composite Optimization.
Lam M. Nguyen, Katya Scheinberg, Trang H. Tran
Journal of Optimization Theory and Applications (JOTA), 2025
- [53] Correlated Attention in Transformers for Multivariate Time Series.
 Quang M. Nguyen, **Lam M. Nguyen**, Subhro Das
The 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025), 2025
- [52] Dilated Convolution for Time Series Learning.
 Wang Zhang, Subhro Das, **Lam M. Nguyen**, Luca Daniel.
The 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025), 2025
- [51] Foundation Model and Temporal Priors-guided Transductive Few-shot Action Recognition.
 Bach Vu, Hoang Nguyen, Quang M. Nguyen, Duong Le, Hieu Pham, Phi Le Nguyen, **Lam M. Nguyen**.
The 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025), 2025
- [50] TabularFM: An Open Framework For Tabular Foundational Models.
 Quan M. Tran, Suong N. Hoang, **Lam M. Nguyen**, Dzung Phan, Hoang Thanh Lam
2024 IEEE International Conference on Big Data (IEEE BigData 2024), 2024 (19.7% acceptance rate)
- [49] Abstracted Shapes as Tokens - A Generalizable and Interpretable Model for Time-series Classification.
 Yunshi Wen, Tengfei Ma, Tsui-Wei Weng, **Lam M. Nguyen**, Anak Agung Julius
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024), 2024 (25.8% acceptance rate)
- [48] Shuffling Gradient-Based Methods for Nonconvex-Concave Minimax Optimization.
 Quoc Tran-Dinh, Trang H. Tran, **Lam M. Nguyen**
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024), 2024 (25.8% acceptance rate)
- [47] Probabilistic Federated Prompt-Tuning in Data Imbalance Settings.
 Pei-Yau Weng, Minh Hoang, **Lam M. Nguyen**, My T. Thai, Tsui-Wei Weng, Trong Nghia Hoang
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024), 2024 (25.8% acceptance rate)
- [46] Improving Time Series Encoding with Noise-Aware Self-Supervised Learning and an Efficient Encoder.
 Anh Duy Nguyen, Trang H. Tran, Hieu H. Pham, Phi Le Nguyen, **Lam M. Nguyen**
The 24th IEEE International Conference on Data Mining (ICDM 2024), 2024 (19.5% acceptance rate)
- [45] Proactive DP: A Multiple Target Optimization Framework for DP-SGD.
 Marten van Dijk, Nhung Van Nguyen, Toan N. Nguyen, **Lam M. Nguyen**, Phuong Ha Nguyen
The 41th International Conference on Machine Learning (ICML 2024), 2024 (27.5% acceptance rate)
- [44] Shuffling Momentum Gradient Algorithm for Convex Optimization.
 Trang H. Tran, Quoc Tran-Dinh, **Lam M. Nguyen**
Vietnam Journal of Mathematics (VJOM), Special issue dedicated to Dr. Tamás Terlaky on the occasion of his 70th birthday, 2024
- [43] Multi-polytope Machine for Classification.
 Dzung Phan, **Lam M. Nguyen**, Jayant Kalagnanam, Chandra Reddy
SIAM Conference on Data Mining (SDM 2024), 2024 (29.2% acceptance rate)

- [42] On Partial Optimal Transport: Revising the Infeasibility of Sinkhorn and Efficient Gradient Methods.
Anh Duc Nguyen, Tuan Dung Nguyen, Quang Nguyen, Hoang Nguyen, **Lam M. Nguyen**, Kim-Chuan Toh
The 38th AAAI Conference on Artificial Intelligence (AAAI 2024), 2024 (23.75% acceptance rate)
INFORMS 2024 Undergraduate Operations Research Prize
- [41] One Step Closer to Unbiased Aleatoric Uncertainty Estimation.
Wang Zhang, Martin Ma, Subhro Das, Lily Weng, Alexandre Megretsky, Luca Daniel, **Lam M. Nguyen**
The 38th AAAI Conference on Artificial Intelligence (AAAI 2024), 2024 (23.75% acceptance rate)
- [40] On Unbalanced Optimal Transport: Gradient Methods, Sparsity and Approximation Error.
Quang Minh Nguyen, Hoang H. Nguyen, Yi Zhou, **Lam M. Nguyen**
Journal of Machine Learning Research (JMLR), 2023
- [39] On the Convergence to a Global Solution of Shuffling-Type Gradient Algorithms.
Lam M. Nguyen, Trang H. Tran
The 37th Conference on Neural Information Processing Systems (NeurIPS 2023), 2023 (26.1% acceptance rate)
- [38] Analyzing Generalization of Neural Networks through Loss Path Kernels.
Yilan Chen, Wei Huang, Hao Wang, Charlotte Loh, Akash Srivastava, **Lam M. Nguyen**, Tsui-Wei Weng
The 37th Conference on Neural Information Processing Systems (NeurIPS 2023), 2023 (26.1% acceptance rate)
- [37] Attacking c-MARL More Effectively: A Data Driven Approach.
Nhan Pham, **Lam M. Nguyen**, Jie Chen, Hoang Thanh Lam, Subhro Das, and Tsui-Wei Weng
The 23rd IEEE International Conference on Data Mining (ICDM 2023), 2023 (19.94% acceptance rate)
- [36] Promoting Robustness of Randomized Smoothing: Two Cost-Effective Approaches.
Linbo Liu, Trong Nghia Hoang, **Lam M. Nguyen**, and Tsui-Wei Weng
The 23rd IEEE International Conference on Data Mining (ICDM 2023), 2023 (19.94% acceptance rate)
- [35] ConCerNet: A Contrastive Learning Based Framework for Automated Conservation Law Discovery and Trustworthy Dynamical System Prediction.
Wang Zhang, Tsui-Wei Weng, Subhro Das, Alexandre Megretsky, Luca Daniel, **Lam M. Nguyen**
The 40th International Conference on Machine Learning (ICML 2023), 2023 (27.9% acceptance rate)
- [34] Scalable and Secure Federated XGBoost.
Quang Nguyen, Nhan Khanh Le, **Lam M. Nguyen**.
The 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2023), 2023
- [33] Label-free Concept Bottleneck Models.
Tuomas Oikarinen, Subhro Das, **Lam M. Nguyen**, Tsui-Wei Weng.
The 11th International Conference on Learning Representations (ICLR 2023), 2023
- [32] Optimal Control via Linearizable Deep Learning.
Vinicius Lima, Dzung T. Phan, **Lam M. Nguyen**, Jayant R. Kalagnanam
The 2023 American Control Conference (ACC 2023), 2023
- [31] Nesterov Accelerated Shuffling Gradient Method for Convex Optimization.
Trang H. Tran, Katya Scheinberg, **Lam M. Nguyen**
The 39th International Conference on Machine Learning (ICML 2022), 2022 (21.9% acceptance rate)

- [30] Finite-Sum Smooth Optimization with SARAH.
Lam M. Nguyen, Marten van Dijk, Dzung T. Phan, Phuong Ha Nguyen, Tsui-Wei Weng, Jayant R. Kalagnanam
Computational Optimization and Applications (COAP), 2022
- [29] AI-based Real-time Site-wide Optimization for Process Manufacturing.
Jayant Kalagnanam, Dzung Phan, Pavankumar Murali, **Lam M. Nguyen**, Nianjun Zhou, Dharmashankar Subramanian, Raju Pavuluri, Xiang Ma, Crystal Lui, Giovane Cesar da Silva
INFORMS Journal on Applied Analytics (IJAA), 2022
- [28] StepDIRECT - A Derivative-Free Optimization Method for Stepwise Functions.
Dzung Phan, Hongsheng Liu, **Lam M. Nguyen**
SIAM International Conference on Data Mining (SDM22), 2022 (27.8% acceptance rate)
- [27] Besting the Black-Box: Barrier Zones for Adversarial Example Defense.
Kaleel Mahmood, Phuong Ha Nguyen, **Lam M. Nguyen**, Thanh Nguyen, Marten van Dijk
IEEE Access, 2022
- [26] Interpretable Clustering via Multi-Polytope Machines.
Connor Lawless, Jayant Kalagnanam, **Lam M. Nguyen**, Dzung Phan, Chandra Reddy
The 36th AAAI Conference on Artificial Intelligence (AAAI 2022), 2022 (15% acceptance rate)
- [25] FedDR - Randomized Douglas-Rachford Splitting Algorithms for Nonconvex Federated Composite Optimization.
Quoc Tran-Dinh, Nhan Pham, Dzung T. Phan, **Lam M. Nguyen**
The 35th Conference on Neural Information Processing Systems (NeurIPS 2021), 2021 (26% acceptance rate)
- [24] Ensembling Graph Predictions for AMR Parsing.
Thanh Lam Hoang, Gabriele Picco, Yufang Hou, Young-Suk Lee, **Lam M. Nguyen**, Dzung T. Phan, Vanessa López, Ramon Fernandez Astudillo
The 35th Conference on Neural Information Processing Systems (NeurIPS 2021), 2021 (26% acceptance rate)
- [23] On the Equivalence between Neural Network and Support Vector Machine.
Yilan Chen, Wei Huang, **Lam M. Nguyen**, Tsui-Wei Weng
The 35th Conference on Neural Information Processing Systems (NeurIPS 2021), 2021 (26% acceptance rate)
- [22] A Unified Convergence Analysis for Shuffling-Type Gradient Methods.
Lam M. Nguyen, Quoc Tran-Dinh, Dzung T. Phan, Phuong Ha Nguyen, Marten van Dijk
Journal of Machine Learning Research (JMLR), volume 22, 1-43, 2021
- [21] SMG: A Shuffling Gradient-Based Method with Momentum.
Trang H. Tran, **Lam M. Nguyen**, Quoc Tran-Dinh
The 38th International Conference on Machine Learning (ICML 2021), PMLR 139, 2021 (21.47% acceptance rate)
- [20] Regression Optimization for System-level Production Control.
Dzung T. Phan, **Lam M. Nguyen**, Pavankumar Murali, Nhan H. Pham, Hongsheng Liu, Jayant R. Kalagnanam
The 2021 American Control Conference (ACC 2021), 2021
- [19] Hogwild! over Distributed Local Data Sets with Linearly Increasing Mini-Batch Sizes.
Nhuong V. Nguyen, Toan N. Nguyen, Phuong Ha Nguyen, Quoc Tran-Dinh, **Lam M. Nguyen**, Marten van Dijk
The 24th International Conference on Artificial Intelligence and Statistics (AISTATS 2021), 2021 (29.8% acceptance rate)

- [18] A Hybrid Stochastic Optimization Framework for Stochastic Composite Nonconvex Optimization.
Quoc Tran-Dinh, Nhan H. Pham, Dzung T. Phan, **Lam M. Nguyen**
Mathematical Programming (MAPR), 2021
IBM 2022 Pat Goldberg Memorial Best Paper Award
- [17] Hybrid Variance-Reduced SGD Algorithms for Nonconvex-Concave Minimax Problems.
Quoc Tran-Dinh, Deyi Liu, **Lam M. Nguyen**
The 34th Conference on Neural Information Processing Systems (NeurIPS 2020), 2020
(20.1% acceptance rate)
- [16] A Scalable MIP-based Method for Learning Optimal Multivariate Decision Trees.
Haoran Zhu, Pavankumar Murali, Dzung T. Phan, **Lam M. Nguyen**, Jayant R. Kalagnanam
The 34th Conference on Neural Information Processing Systems (NeurIPS 2020), 2020
(20.1% acceptance rate)
- [15] Inexact SARAH Algorithm for Stochastic Optimization.
Lam M. Nguyen, Katya Scheinberg, Martin Takac
Optimization Methods and Software (GOMS), volume 36(1), 237-258, 2020
- [14] Pruning Deep Neural Networks with L0-constrained Optimization.
Dzung T. Phan, **Lam M. Nguyen**, Nam H. Nguyen, Jayant R. Kalagnanam
The 20th IEEE International Conference on Data Mining (ICDM 2020), 2020 (19.7% acceptance rate)
- [13] Stochastic Gauss-Newton Algorithms for Nonconvex Compositional Optimization.
Quoc Tran-Dinh, Nhan H. Pham, **Lam M. Nguyen**
The 37th International Conference on Machine Learning (ICML 2020), PMLR 119, 2020 (21.8% acceptance rate)
- [12] ProxSARAH: An Efficient Algorithmic Framework for Stochastic Composite Nonconvex Optimization.
Nhan H. Pham, **Lam M. Nguyen**, Dzung T. Phan, Quoc Tran-Dinh
Journal of Machine Learning Research (JMLR), volume 21(110), 1-48, 2020
IBM 2020 Pat Goldberg Memorial Best Paper Competition - Finalist
- [11] A Hybrid Stochastic Policy Gradient Algorithm for Reinforcement Learning.
Nhan H. Pham, **Lam M. Nguyen**, Dzung T. Phan, Phuong Ha Nguyen, Marten van Dijk, Quoc Tran-Dinh
The 23rd International Conference on Artificial Intelligence and Statistics (AISTATS 2020), PMLR 108, 2020
- [10] New Convergence Aspects of Stochastic Gradient Algorithms.
Lam M. Nguyen, Phuong Ha Nguyen, Peter Richtarik, Katya Scheinberg, Martin Takac, Marten van Dijk
Journal of Machine Learning Research (JMLR), volume 20(176), 1-49, 2019
- [9] Tight Dimension Independent Lower Bound on the Expected Convergence Rate for Diminishing Step Sizes in SGD.
Phuong Ha Nguyen, **Lam M. Nguyen**, Marten van Dijk
The 33th Conference on Neural Information Processing Systems (NeurIPS 2019), 2019
(21.17% acceptance rate)
- [8] PROVEN: Verifying Robustness of Neural Networks with a Probabilistic Approach.
Tsui-Wei Weng, Pin-Yu Chen, **Lam M. Nguyen**, Mark S. Squillante, Akhilan Boopathy, Ivan Oseledets, Luca Daniel
The 36th International Conference on Machine Learning (ICML 2019), PMLR 97, 2019 (22.5% acceptance rate)
- [7] Characterization of Convex Objective Functions and Optimal Expected Convergence Rates for SGD.
Marten van Dijk, **Lam M. Nguyen**, Phuong Ha Nguyen, Dzung T. Phan
The 36th International Conference on Machine Learning (ICML 2019), PMLR 97, 2019 (22.5% acceptance rate)

- [6] Chief: A Change Pattern based Interpretable Failure Analyzer.
Dhaval Patel, **Lam M. Nguyen**, Akshay Rangamani, Shrey Shrivastava, Jayant Kalagnanam
2018 IEEE International Conference on Big Data (IEEE BigData 2018), 2018
- [5] SGD and Hogwild! Convergence Without the Bounded Gradients Assumption.
Lam M. Nguyen, Phuong Ha Nguyen, Marten van Dijk, Peter Richtarik, Katya Scheinberg, Martin Takac
The 35th International Conference on Machine Learning (ICML 2018), PMLR 80, 2018 (25% acceptance rate)
IBM Research AI – Selected Publications 2018
- [4] SARAH: A Novel Method for Machine Learning Problems Using Stochastic Recursive Gradient.
Lam M. Nguyen, Jie Liu, Katya Scheinberg, Martin Takac
The 34th International Conference on Machine Learning (ICML 2017), PMLR 70:2613-2621, 2017 (25% acceptance rate)
Van Hoesen Family Best Publication Award
- [3] A Queueing System with On-demand Servers: Local Stability of Fluid Limits.
Lam M. Nguyen, Alexander L. Stolyar
Queueing Systems (QUESTA), 1-26, Springer, 2017
- [2] A Service System with Randomly Behaving On-demand Agents.
Lam M. Nguyen, Alexander L. Stolyar
The 42nd International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS 2016), ACM SIGMETRICS Performance Evaluation Review, 44(1):365-366, 2016 (25% acceptance rate)
- [1] CEO Compensation: Does Financial Crisis Matter?
Prasad Vemala, **Lam Nguyen**, Dung Nguyen, Alekhya Kommasani
International Business Research, 7(4):125-131, 2014

PEER-REVIEWED WORKSHOP PAPERS

- [8] Guaranteeing Conservation Laws with Projection in Physics-Informed Neural Networks.
Anthony Baez, Wang Zhang, Ziwen Ma, Subhro Das, **Lam M. Nguyen**, Luca Daniel
The 38th Conference on Neural Information Processing Systems (NeurIPS 2024), Data-driven and Differentiable Simulations, Surrogates, and Solvers (D3S3), 2024
- [7] Stochastic FISTA Step Search Algorithm for Convex Optimization.
Trang H. Tran, **Lam M. Nguyen**, Katya Scheinberg
The 37th Conference on Neural Information Processing Systems (NeurIPS 2023), Optimization for Machine Learning (OPT 2023), 2023
- [6] c-MBA: Adversarial Attack for Cooperative MARL Using Learned Dynamics Model.
Nhan H. Pham, **Lam M. Nguyen**, Jie Chen, Hoang Thanh Lam, Subhro Das, Tsui-Wei Weng
The 36th Conference on Neural Information Processing Systems (NeurIPS 2022), ML Safety, 2022
- [5] Fast Convergence for Unstable Reinforcement Learning Problems by Logarithmic Mapping.
Wang Zhang, **Lam M. Nguyen**, Subhro Das, Alexandre Megretski, Luca Daniel, Tsui-Wei Weng
The 39th International Conference on Machine Learning (ICML 2022), Decision Awareness in Reinforcement Learning, 2022
- [4] Robust Randomized Smoothing via Two Cost-Effective Approaches.
Linbo Liu, Trong Nghia Hoang, **Lam M. Nguyen**, Tsui-Wei Weng
The 10th International Conference on Learning Representations (ICLR 2022), PAIR2Struct: Privacy, Accountability, Interpretability, Robustness, Reasoning on Structured Data, 2022

- [3] Addressing Solution Quality in Data Generated Optimization Models.
Orit Davidovich, Parikshit Ram, Segev Wasserkrug, Dharmashankar Subramanian, Nianjun Zhou, Dzung Phan, Pavankumar Murali, **Lam M. Nguyen**
The 36th AAAI Conference on Artificial Intelligence (AAAI 2022), AI for Decision Optimization, AI4DO, 2022
- [2] Automated Decision Optimization: Data and Knowledge Driven Optimization Model Generation with Human-in-the-loop.
Lisa Amini, Arunima Chaudhary, Yishai Feldman, Pavankumar Murali, **Lam M. Nguyen**, Dzung Phan, Aviad Sela, Carolina Spina, Dharmashankar Subramanian, Abel Valente, Long Vu, Dakuo Wang, Segev Wasserkrug, Ritesh Yadav, Nianjun Zhou
The 36th AAAI Conference on Artificial Intelligence (AAAI 2022), AI for Decision Optimization, AI4DO, 2022
- [1] Ensuring the Quality of Optimization Solutions in Data Generated Optimization Models.
Segev Wasserkrug, Orit Davidovich, Evgeny Shindin, Dharmashankar Subramanian, Parikshit Ram, Pavankumar Murali, Dzung Phan, Nianjun Zhou, **Lam M. Nguyen**
The 30th International Joint Conference on Artificial Intelligence (IJCAI 2021), Data Science Meets Optimisation, DSO@IJCAI2021, 2021

PREPRINTS

- [15] Learning to Shuffle: Block Reshuffling and Reversal Schemes for Stochastic Optimization.
Lam M. Nguyen, Dzung T. Phan, Jayant Kalagnanam
Technical report, arXiv preprint, 2026
- [14] A Supervised Contrastive Learning Pretrain-Finetune Approach for Time Series.
Trang H. Tran, **Lam M. Nguyen**, Kyongmin Yeo, Nam Nguyen, Roman Vaculin
Technical report, arXiv preprint, 2023
- [13] Batch Clipping and Adaptive Layerwise Clipping for Differential Private Stochastic Gradient Descent.
Toan N. Nguyen, Phuong Ha Nguyen, **Lam M. Nguyen**, Marten van Dijk
Technical report, arXiv preprint, 2023
- [12] An End-to-End Time Series Model for Simultaneous Imputation and Forecast.
Trang H. Tran, **Lam M. Nguyen**, Kyongmin Yeo, Nam Nguyen, Dzung Phan, Roman Vaculin, Jayant Kalagnanam
Technical report, arXiv preprint, 2023
- [11] Generalizing DP-SGD with Shuffling and Batching Clipping.
Marten van Dijk, Phuong Ha Nguyen, Toan N. Nguyen, **Lam M. Nguyen**
Technical report, arXiv preprint, 2022
- [10] Finding Optimal Policy for Queueing Models: New Parameterization.
Trang H. Tran, **Lam M. Nguyen**, Katya Scheinberg
Technical report, arXiv preprint, 2022
- [9] Finite-Sum Optimization: A New Perspective for Convergence to a Global Solution.
Lam M. Nguyen, Trang H. Tran, Marten van Dijk
Technical report, arXiv preprint, 2022
- [8] Differential Private Hogwild! over Distributed Local Data Sets.
Marten van Dijk, Nhung V. Nguyen, Toan N. Nguyen, **Lam M. Nguyen**, Phuong Ha Nguyen
Technical report, arXiv preprint, 2021
- [7] An Optimal Hybrid Variance-Reduced Algorithm for Stochastic Composite Nonconvex Optimization.
Deyi Liu, **Lam M. Nguyen**, Quoc Tran-Dinh
Technical report, arXiv preprint, 2020

- [6] Asynchronous Federated Learning with Reduced Number of Rounds and with Differential Privacy from Less Aggregated Gaussian Noise.
Marten van Dijk, Nhuong V. Nguyen, Toan N. Nguyen, **Lam M. Nguyen**, Quoc Tran-Dinh, Phuong Ha Nguyen
Technical report, arXiv preprint, 2020
- [5] Finite-Time Analysis of Stochastic Gradient Descent under Markov Randomness.
Thinh T. Doan, **Lam M. Nguyen**, Nhan H. Pham, Justin Romberg
Technical report, arXiv preprint, 2020
- [4] Convergence Rates of Accelerated Markov Gradient Descent with Applications in Reinforcement Learning.
Thinh T. Doan, **Lam M. Nguyen**, Nhan H. Pham, Justin Romberg
Technical report, arXiv preprint, 2020
- [3] Hybrid Stochastic Gradient Descent Algorithms for Stochastic Nonconvex Optimization.
Quoc Tran-Dinh, Nhan H. Pham, Dzung T. Phan, **Lam M. Nguyen**
Technical report, arXiv preprint, 2019
- [2] When Does Stochastic Gradient Algorithm Work Well?
Lam M. Nguyen, Nam H. Nguyen, Dzung T. Phan, Jayant R. Kalagnanam, Katya Scheinberg
Technical report, arXiv preprint, 2018
- [1] Stochastic Recursive Gradient Algorithm for Nonconvex Optimization.
Lam M. Nguyen, Jie Liu, Katya Scheinberg, Martin Takac
Technical report, arXiv preprint, 2017

GRANTED PATENTS

- [23] A Neural Network Based Dynamical System Modeling Method to Ensure The Contrastively Learned Conservation Laws. Patent 12743634
Lam M. Nguyen, Wang Zhang, Subhro Das, Alexandre Megretski, Luca Daniel
- [22] Evolution and Regression Generative Models and Sequence Representation Learning from Multi Sequence Alignment and Phylogenetic Trees Data. Patent 12725679
Thanh Lam Hoang, Marcos Martínez Galindo, Gabriele Picco, Mykhaylo Zayats, Nhan Huu Pham, **Lam M. Nguyen**, Marco Luca Sbodio, Dzung Tien Phan, Vanessa Lopez Garcia
- [21] An End-to-End Model for Training Decision Trees with Dimension Reduction. Patent 12718113
Dzung T. Phan, Michael Huang, Pavankumar Murali, **Lam M. Nguyen**
- [20] Multi-Polytope Machine for Classification. Patent 12586002
Dzung T. Phan, **Lam M. Nguyen**, Jayant R. Kalagnanam, Chandrasekhara K. Reddy, Srideepika Jayaraman
- [19] Training A Neural Network Using an Accelerated Gradient with Shuffling. Patent 12579434
Lam M. Nguyen, Trang H. Tran
- [18] Certification-based Robust Training by Refining Decision Boundary. Patent 12567232
Lam M. Nguyen, Wang Zhang, Subhro Das, Pin-Yu Chen, Alexandre Megretski, Luca Daniel
- [17] Site-Wide Optimization for Mixed Regression Models and Mixed Control Variables. Patent US12518174B2
Dung Tien Phan, Nhan H. Pham, **Lam M. Nguyen**
- [16] Optimizer Agnostic Explanation System for Large Scale Schedules. Patent US12443454B2
Surya Shravan Kumar Sajja, Kanthi Sarpatwar, **Lam M. Nguyen**, Yuan Yuan Jia, Stephane Michel, Roman Vaculin
- [15] Unsupervised Learning from Public Tabular Datasets. Patent US12436924B2
Thanh Lam Hoang, Gabriele Picco, **Lam M. Nguyen**, Dzung Tien Phan

- [14] System-level Control using Tree-based Regression with Outlier Removal. Patent US20220058515A1
Dung Tien Phan, Pavankumar Murali, **Lam M. Nguyen**
- [13] Automated Generation of Optimization Model for System-Wide Plant Optimization. Patent US20220027685A1
Dung Tien Phan, **Lam M. Nguyen**, Pavankumar Murali, Nianjun Zhou
- [12] Providing a Semantic Encoding and Language Neural Network. Patent US12217007B2
Thanh Lam Hoang, Dzung Phan, Gabriele Picco, **Lam M. Nguyen**, Marco Luca Sbodio, Vanessa Lopez Garcia
- [11] Blending Graph Predictions. Patent US12242801B2
Thanh Lam Hoang, Gabriele Picco, Yufang Hou, Young-Suk Lee, **Lam M. Nguyen**, Dzung Tien Phan, Vanessa Lopez Garcia, Ramon Fernandez Astudillo
- [10] Reasonable Language Model Learning for Text Generation from a Knowledge Graph. Patent US12216996B2
Hoang Thanh Lam, Dzung T. Phan, Gabriele Picco, **Lam M. Nguyen**, Vanessa Lopez Garcia
- [9] Intelligent Dynamic Condition-based Infrastructure Maintenance Scheduling. Patent US12158797B2
Pavankumar Murali, Dzung Tien Phan, Nianjun Zhou, **Lam M. Nguyen**
- [8] Regression-Optimization Control of Production Process with Dynamic Inputs. Patent US20240377810A1
Lam M. Nguyen, Pavankumar Murali, Nianjun Zhou, Binny Winston Samuel
- [7] Compression of Deep Neural Networks. US Patent US20200293876A1
Dzung T. Phan, **Lam M. Nguyen**, Nam H. Nguyen, Jayant R. Kalagnanam
- [6] Federated Learning for Training Machine Learning Models. Patent US20230196081A1
Lam M. Nguyen, Dung Tien Phan, Jayant R. Kalagnanam
- [5] A Method for Tuning Hyper-Parameters for Classification. Patent US20220027757A1
Dung Tien Phan, Hongsheng Liu, **Lam M. Nguyen**
- [4] Optimal Interpretable Decision Trees Using Integer Linear Programming Techniques. Patent US20210264290A1
Pavankumar Murali, Haoran Zhu, Dung Tien Phan, **Lam M. Nguyen**
- [3] Site-wide Operations Management Optimization for Manufacturing and Processing Control. Patent US20220057786A1
Dung Tien Phan, **Lam M. Nguyen**, Pavankumar Murali, and Hongsheng Liu
- [2] A Shuffling-Type Gradient Method for Training Machine Learning models with Big Data. Patent US20220171996A1
Lam M. Nguyen, Dung Tien Phan
- [1] Prediction Optimization for System-level Production Control. Patent US11099529B2
Dzung T. Phan, **Lam M. Nguyen**, Pavankumar Murali, Jayant R. Kalagnanam

PATENTS APPLICATIONS

- [20] Expert-Fusion Time Series Forecasting. Filed on August 21, 2026
Kyong Min Yeo, Chandrasekhara K. Reddy, Wesley M. Gifford, **Lam Minh Nguyen**, Kanthi Sarpatwar, Yating Zhou
- [19] Encoder-Decoder Architecture for Zero-Shot Modeling and Analysis of Time Series Data. Filed on February 17, 2026
Yunshi Wen, **Lam M. Nguyen**, Chandrasekhara K. Reddy, Wesley M. Gifford, Kyong Min Yeo, Kanthi Sarpatwar, Jayant R. Kalagnanam, Anak Agung Julius

- [18] Time-Series Modeling Using Function-to-Function Mapping. Filed on February 06, 2026
Kyong Min Yeo, Chandrasekhara K. Reddy, Wesley M. Gifford, Kanthi Sarpatwar, **Lam M. Nguyen**, Chandni Nagda
- [17] Tokenizer-Predictor Architecture Designed for Zero-Shot Modeling of Time Series Data. Filed on January 30, 2026
Yunshi Wen, **Lam M. Nguyen**, Chandrasekhara K. Reddy, Wesley M. Gifford, Jayant R. Kalagnanam, Agung Julius
- [16] Deep Learning Architecture For Multivariate Time Series. Filed on December 14, 2024
Quang Minh Nguyen, **Lam M. Nguyen**, Subhro Das
- [15] Generative Schrodinger Network. Filed on March 11, 2024
Thanh Lam Hoang, Marcos Martínez Galindo, **Lam M. Nguyen**, Niall Robertson, Vanessa Lopez Garcia
- [14] Process Control Based on Simultaneous Machine Learning of Spatial and Temporal Relations of Time Series Data. Filed on January 05, 2024
Lam M. Nguyen, Trang H. Tran
- [13] Time Series Forecasting Using Multivariate Time Series Data with Missing Values. Filed on January 28, 2023
Lam M. Nguyen, Trang H. Tran, Kyong Min Yeo, Nam H. Nguyen, Dzung Tien Phan, Roman Vaculin, Jayant R. Kalagnanam
- [12] Multivariable Time-Series Feature Extraction. Filed on January 27, 2023
Lam M. Nguyen, Wang Zhang, Subhro Das, Alexandre Megretski, Luca Daniel
- [11] Privacy Enhanced Machine Learning over Graph Data. Filed on January 23, 2023
Ambrish Rawat, Naoise Holohan, Heiko H. Ludwig, Ehsan Degan, Nathalie Baracaldo Angel, Alan Jonathan King, Swanand Ravindra Kadhe, Yi Zhou, Keith Coleman Houck, Mark Purcell, Giulio Zizzo, Nir Drucker, Hayim Shaul, Eyal Kushnir, **Lam M. Nguyen**
- [10] Providing Trained Reinforcement Learning Systems. Filed on December 12, 2022
Lam M. Nguyen, Wang Zhang, Subhro Das, Alexandre Megretski, Luca Daniel
- [9] Active Learning in Model Training. Filed on November 22, 2022
Dzung Tien Phan, Huozhi Zhou, **Lam M. Nguyen**, Chandrasekhara K. Reddy, Jayant R. Kalagnanam
- [8] Automated Decision Optimization for Maintenance of Physical Assets. Filed on October 31, 2022
Nianjun Zhou, Pavankumar Murali, Dzung T. Phan, **Lam M. Nguyen**
- [7] Adversarial Attacks for Improving Cooperative Multi-Agent Reinforcement Learning Systems. Filed on September 23, 2022
Nhan Huu Pham, **Lam M. Nguyen**, Jie Chen, Thanh Lam Hoang, Subhro Das
- [6] Training Neural Networks with Convergence to a Global Minimum. Filed on September 23, 2022
Lam M. Nguyen
- [5] Machine Learning-based Decision Framework for Physical Systems. Filed on September 20, 2022
Dzung Tien Phan, **Lam M. Nguyen**
- [4] System and Method for unsupervised Learning of Semantic Graph from textual data and language generation from Semantic graphH via Reinforcement learning. Filed on July 11, 2022
Thanh Lam Hoang, Dzung Tien Phan, Gabriele Picco, **Lam M. Nguyen**, Marco Luca Sbodio, Vanessa Lopez Garcia
- [3] Integrated Machine Learning Prediction and Optimization for Decision-Making. Filed on March 30, 2022
Dzung T. Phan, Long Vu, **Lam M. Nguyen**, Dharmashankar Subramanian

- [2] Interpretable Clustering via Multi-Polytope Machines. Filed on February 18, 2022
 Dzung T. Phan, Connor Lawless, Jayant R. Kalagnanam, **Lam M. Nguyen**, Chandrasekhara K. Reddy
- [1] Optimal Control of Dynamic Systems via Linearizable Deep Learning. Filed on February 07, 2022
 Dung Tien Phan, Jayant R. Kalagnanam, **Lam M. Nguyen**

THESES

- 2018 A Service System with On-Demand Agents, Stochastic Gradient Algorithms and the SARAH Algorithm.
Lam M. Nguyen
PhD dissertation, Lehigh University, Bethlehem, PA
Elizabeth V. Stout Dissertation Award
- 2008 Methods for Detecting Hidden Period in Some Economics Processes.
Lam M. Nguyen
Undergraduate thesis, Lomonosov Moscow State University, Moscow, Russia

ORGANIZING WORKSHOPS

- [3] Principled Design for Trustworthy AI: Interpretability, Robustness, and Safety Across Modalities.
 Tsui-Wei Weng, Trong Nghia Hoang, Tengfei Ma, Jake C. Snell, Francesco Croce, Chandan Singh, Subarna Tripathi, **Lam M. Nguyen**
Workshop at The 14th International Conference on Learning Representations (ICLR 2026), 2026
- [2] When Machine Learning meets Dynamical Systems: Theory and Applications.
Lam M. Nguyen, Trang H. Tran, Wang Zhang, Subhro Das, Tsui-Wei Weng
Workshop at The 37th Conference on Artificial Intelligence (AAAI 2023), 2023
- [1] New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership.
 Nghia Hoang, **Lam M. Nguyen**, Pin-Yu Chen, Tsui-Wei Weng, Sara Magliacane, Bryan Kian Hsiang Low, Anoop Deoras
Workshop at The 35th Conference on Neural Information Processing Systems (NeurIPS 2021), 2021

INVITED TALKS

- 10/2023 On the Convergence to a Global Solution of Shuffling-Type Gradient Algorithms.
INFORMS Annual Meeting, Phoenix, AZ
- 10/2022 New Perspective On The Convergence To A Global Solution Of Finite-sum Optimization.
INFORMS Annual Meeting, Indianapolis, IN
- 09/2022 Nesterov Accelerated Shuffling Gradient Method for Convex Optimization.
Johns Hopkins University, Baltimore, MD
- 10/2021 Hogwild! Over Distributed Local Data Sets With Linearly Increasing Mini-batch Sizes.
INFORMS Annual Meeting, Anaheim, CA
- 11/2020 A Unified Convergence Analysis for Shuffling-Type Gradient Methods.
INFORMS Annual Meeting, Virtual Conference
- 10/2019 Finite-Sum Smooth Optimization with SARAH.
INFORMS Annual Meeting, Seattle, WA
- 11/2018 Inexact SARAH for Solving Stochastic Optimization Problems.
INFORMS Annual Meeting, Phoenix, AZ
- 08/2018 Inexact SARAH for Solving Stochastic Optimization Problems.
DIMACS/TRIPODS/MOPTA, Bethlehem, PA
- 03/2018 When does stochastic gradient algorithm work well?
INFORMS Optimization Society Conference, Denver, CO

10/2017	SARAH: Stochastic Recursive Gradient Algorithm. <i>INFORMS Annual Meeting</i> , Houston, TX
08/2017	SARAH Algorithm. <i>IBM Thomas J. Watson Research Center</i> , Yorktown Heights, NY
11/2016	A Queueing System with On-demand Servers: Local Stability of Fluid Limits. <i>INFORMS Annual Meeting</i> , Nashville, TN
08/2016	A Queueing System with On-demand Servers: Local Stability of Fluid Limits. <i>Modeling and Optimization: Theory and Applications</i> , Bethlehem, PA

PROFESSIONAL ACTIVITIES

MEMBER

2025 – 2026	Scientific Board Member , Vietnam AI Championship (VAIC)
03/2024 – Present	Senior Member , Institute for Operations Research and the Management Sciences (INFORMS)
06/2020 – Present	Editorial Board , Journal of Machine Learning Research
06/2021 – Present	Editorial Board , Machine Learning
06/2022 – Present	Editorial Board , Journal of Optimization Theory and Applications
01/2022 – 12/2023	Editorial Board , IEEE Transactions on Neural Networks and Learning Systems
01/2022 – 12/2022	Editorial Board , Neural Networks
2023	Program Committee , “ <i>When Machine Learning meets Dynamical Systems: Theory and Applications (MLmDS 2023)</i> ”, AAAI 2023 Workshop
2021	Program Committee , “ <i>New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership (NFFL 2021)</i> ”, NeurIPS 2021 Workshop
2020	Program Committee , “ <i>Optimization for Machine Learning (OPT 2020)</i> ”, NeurIPS 2020 Workshop
2018	Program Committee , “ <i>Modern Trends in Nonconvex Optimization for Machine Learning</i> ”, ICML 2018 Workshop

REVIEWER / PROGRAM COMMITTEE (PEER-REVIEWED CONFERENCES)

2017 – 2019	International Conference on Machine Learning (ICML)
2017 – 2021	Conference on Neural Information Processing Systems (NIPS/NeurIPS)
2018 – 2020	International Conference on Learning Representations (ICLR)
2019 – 2020	International Conference on Artificial Intelligence and Statistics (AISTATS)
2021 – 2022	Conference on Learning Theory (COLT)
2019 – 2021	AAAI Conference on Artificial Intelligence (AAAI)
2020	International Joint Conferences on Artificial Intelligence (IJCAI)
2019 – 2022	IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
2019 – 2021	IEEE International Conference on Computer Vision (ICCV)
2020	European Conference on Computer Vision (ECCV)
2019 – 2021	Conference on Uncertainty in Artificial Intelligence (UAI)

REVIEWER (PEER-REVIEWED JOURNALS)

2018 – 2022	Journal of Machine Learning Research
2020 – 2022	Mathematical Programming
2020 – 2021	SIAM Journal on Optimization
2021	SIAM Journal on Numerical Analysis
2020 – 2021	IEEE Transactions on Neural Networks and Learning Systems
2019 – 2020	IEEE Transactions on Signal Processing
2019	Artificial Intelligence

2018	Optimization Methods and Software
2020	SIAM Journal on Mathematics of Data Science
SESSION CHAIR / ORGANIZER (CONFERENCES)	
	International Conference on Machine Learning (ICML)
2022	- Sessions “ <i>OPT: Non-Convex</i> ” and “ <i>Optimization/Reinforcement Learning</i> ”
2021	- Sessions “ <i>Optimization (Stochastic)</i> ” and “ <i>Optimization (Nonconvex)</i> ”
	International Conference on Learning Representations (ICLR)
2021	- Session “ <i>Oral Session 6</i> ”
	International Conference on Artificial Intelligence and Statistics (AISTATS)
2021	- Session “ <i>Theory and Practice of Machine Learning</i> ”
	INFORMS Annual Meeting
2023	- Session “ <i>First-Order Methods for Machine Learning</i> ”
2022	- Session “ <i>Optimization for Machine Learning</i> ”
2021	- Session “ <i>Recent Advances in Stochastic Gradient Algorithms</i> ”
2020	- Session “ <i>Recent Advances in Stochastic Gradient Algorithms for Machine Learning</i> ”
2019	- Session “ <i>Fast and Provable Nonconvex Optimization Algorithms in Machine Learning</i> ”
2018	- Session “ <i>Recent Advances in Optimization Methods for Machine Learning</i> ”
	DIMACS/TRIPODS/MOPTA
2018	- Sessions “ <i>Sparse Optimization</i> ” and “ <i>Stochastic Gradient Descent</i> ”
IBM ACTIVITIES	
06/2026 – Present	Chair , Invention Development Team (IDT)
09/2024 – 06/2026	Co-Chair , Invention Development Team (IDT)
01/2022 – 12/2025	Champion , International Conference on Machine Learning (ICML)
11/2021 – Present	Member , Invention Development Team (IDT)
07/2021 – 12/2024	Champion , Professional Interest Community (PIC) - Learning
09/2020 – 12/2023	Organizer , Department Seminar Series
2022	Member , Research AI Pillar Accomplishment Committee
2021 - 2023	Reviewer , Pat Goldberg Memorial Best Paper Competition
2020	Reviewer , IBM Ph.D. Fellowships
SOCIETY MEMBERSHIPS	
2023 – Present	INFORMS Optimization Society
2022 – Present	Association for the Advancement of Artificial Intelligence (AAAI)
2016 – Present	Society for Industrial and Applied Mathematics (SIAM)
2014 – Present	The Institute for Operations Research and the Management Sciences (INFORMS)
2014 – Present	Beta Gamma Sigma (The International Business Honor Society)

MENTORSHIP

PH.D. STUDENTS

03/2021 – 02/2025	Wang Zhang , Ph.D. student, Department of Mechanical Engineering, <i>Massachusetts Institute of Technology</i> (co-advise with Prof. Luca Daniel).
10/2019 – 01/2025	Trang H. Tran , Ph.D. student, School of Operations Research and Information Engineering, <i>Cornell University</i> (co-advise with Prof. Katya Scheinberg).
08/2018 – 12/2021	Nhan H. Pham , Ph.D. student, Department of Statistics and Operations Research, <i>University of North Carolina at Chapel Hill</i> (co-advise with Prof. Quoc Tran-Dinh). Now at <i>IBM Research</i> , USA.

IBM PH.D. FELLOWSHIP STUDENTS

09/2026 – 06/2027 **Hoang H. Nguyen**, Ph.D. student, H. Milton Stewart School of Industrial and Systems Engineering, *Georgia Institute of Technology*.

IBM RESEARCH INTERNS

05/2026 – 08/2026 **Nicolas Lizarralde**, Ph.D. Student, Department of Electrical, Computer, and Systems Engineering, *Rensselaer Polytechnic Institute*.

05/2025 – 08/2025 **Chandni Nagda**, Ph.D. student, Department of Computer Science, *University of Illinois Urbana-Champaign*.

05/2025 – 08/2025 **Yating Zhou**, Ph.D. Student, Department of Electrical, Computer, and Systems Engineering, *Rensselaer Polytechnic Institute*.

05/2025 – 08/2025 **Yunshi Wen**, Ph.D. Student, Department of Electrical, Computer, and Systems Engineering, *Rensselaer Polytechnic Institute*.

06/2023 – 09/2023 **Quang M. Nguyen**, Ph.D. student, Department of Electrical Engineering and Computer Science, *Massachusetts Institute of Technology*.

06/2022 – 09/2022 **Tuomas Oikarinen**, Ph.D. student, Department of Computer Science and Engineering, *University of California San Diego*.

05/2022 – 08/2022 **Vinicius Lima Silva**, Ph.D. student, Department of Electrical and Systems Engineering, *University of Pennsylvania*.

05/2023 – 08/2023, 05/2022 – 08/2022 **Trang H. Tran**, Ph.D. student, School of Operations Research and Information Engineering, *Cornell University*.

05/2021 – 08/2021 **Connor Lawless**, Ph.D. student, School of Operations Research and Information Engineering, *Cornell University*.

05/2021 – 08/2021 **Huozhi Zhou**, Ph.D. student, Department of Electrical and Computer Engineering, *University of Illinois Urbana-Champaign*.

05/2021 – 08/2021 **Nathanael Assefa**, Ph.D. student, Department of Computer Science, *University of Illinois Urbana-Champaign*.

06/2020 – 09/2020 **Michael Huang**, Ph.D. student, Department of Data Science and Operations, Marshall School of Business, *University of Southern California*.

06/2020 – 08/2020 **Nhan H. Pham**, Ph.D. student, Department of Statistics and Operations Research, *University of North Carolina at Chapel Hill* (student of Prof. Quoc Tran-Dinh) (IBM Research Intern). Now at *IBM Research, USA*.

05/2019 – 12/2019 **Hongsheng Liu**, Ph.D. student, Department of Statistics and Operations Research, *University of North Carolina at Chapel Hill*. Now at *Huawei Technologies Co., Ltd., China*.

01/2019 – 08/2019 **Haoran Zhu**, Ph.D. student, Department of Industrial and Systems Engineering, *University of Wisconsin – Madison*. Now at *Microsoft, USA*.

RPI-IBM PROJECT

04/2025 – Present **Nicolas Lizarralde**, Ph.D. Student, Department of Electrical, Computer, and Systems Engineering, *Rensselaer Polytechnic Institute* (student of Prof. Agung Julius).

01/2024 – Present **Yunshi Wen**, Ph.D. Student, Department of Electrical, Computer, and Systems Engineering, *Rensselaer Polytechnic Institute* (student of Prof. Agung Julius).

MIT-IBM PROJECTS

04/2023 – 02/2025 **Martin Ma**, M.E. Student, School of Engineering and Applied Sciences, *Harvard University*.

03/2021 – 02/2025 **Wang Zhang**, Ph.D. student, Department of Mechanical Engineering, *Massachusetts Institute of Technology* (student of Prof. Luca Daniel).

09/2020 – 09/2021 **Vindula Jayawardana**, Ph.D. student, Department of Electrical Engineering and Computer Science, *Massachusetts Institute of Technology* (student of Prof. Cathy Wu).

MIT SUPERUROP UNDERGRADUATE RESEARCH PROGRAM

- 09/2023 – 05/2024 **Anthony Baez**, Undergraduate student, Electrical Engineering and Computer Science, *Massachusetts Institute of Technology* (co-advise with Prof. Luca Daniel).
- 06/2022 – 05/2023 **Angelos Assos**, Undergraduate student, Computer Science and Mathematics, *Massachusetts Institute of Technology* (co-advise with Prof. Luca Daniel).

EXTERNAL STUDENTS

- 04/2024 – Present **Duc Toan Nguyen**, Ph.D. student, Department of Electrical and Computer Engineering, *Rice University*.
- 09/2022 – 12/2024 **Anh Duy Nguyen**, Ph.D. student, Department of Electrical and Computer Engineering, *University of Illinois at Urbana-Champaign*.
- 10/2021 – 12/2022 **Linbo Liu**, Ph.D. student, Department of Mathematics, *University of California San Diego*.
- 06/2021 – Present **Quang M. Nguyen**, Ph.D. student, Department of Electrical Engineering and Computer Science, *Massachusetts Institute of Technology*.
- 06/2021 – Present **Hoang H. Nguyen**, Ph.D. student, H. Milton Stewart School of Industrial and Systems Engineering, *Georgia Institute of Technology*.
- 03/2021 – 12/2022 **Yilan Chen**, Ph.D. student, Department of Computer Science and Engineering, *University of California San Diego* (student of Prof. Tsui-Wei Weng).
- 01/2019 – 12/2022 **Toan N. Nguyen**, Ph.D. student, Department of Computer Science and Engineering, *University of Connecticut* (student of Prof. Marten van Dijk).
- 01/2019 – 11/2021 **Nhuong V. Nguyen**, Ph.D. student, Department of Computer Science and Engineering, *University of Connecticut* (student of Prof. Marten van Dijk).

PH.D. THESIS COMMITTEE MEMBERSHIP

- 10/2025 – 12/2025 **Thi My Le Le**, Ph.D. student, Laboratoire de Conception, Optimisation et Modélisation des Systèmes, *Université de Lorraine* (student of Prof. Le Thi Hoai An).
- 10/2021 – 01/2025 **Trang H. Tran**, Ph.D. student, School of Operations Research and Information Engineering, *Cornell University* (student of Prof. Katya Scheinberg).
- 09/2020 – 06/2022 **Deyi Liu**, Ph.D. student, Department of Statistics and Operations Research, *University of North Carolina at Chapel Hill* (student of Prof. Quoc Tran-Dinh). Now at *Bytedance*, USA.

TEACHING EXPERIENCE

- 08/2024 – 12/2025 **Adjunct Professor**, Department of Industrial and Systems Engineering, *Lehigh University*, Bethlehem, PA
Course: Optimization Methods in Machine Learning (ISE 444) (Fall 2024, Fall 2025)
- 09/2014 – 05/2015 **Teaching Assistant**, Department of Industrial and Systems Engineering, *Lehigh University*, Bethlehem, PA
Courses: Engineering Probability (ISE 111), Applied Engineering Statistics (ISE 121)
- 01/2012 – 12/2013 **Graduate (Teaching) Assistant**, College of Business, *McNeese State University*, Lake Charles, LA
Courses: Human Resource Management (MGMT 310), Staffing (MGMT 315), Strategic Management (MGMT 481), Management Theory and Organizational Behavior (MGMT 604), Issues in Global Business (BADM 218), Entrepreneurial Finance for Small Business (FIN 308)
- 09/2007 – 05/2008 **Teaching Assistant**, Faculty of Computational Mathematics and Cybernetics, *Lomonosov Moscow State University*, Moscow, Russia
Courses: Mathematical Analysis (Calculus), Linear Algebra and Analytic Geometry

OTHER WORK EXPERIENCE

05/2013 – 08/2013 **Graduate Assistant**, College of Business, *McNeese State University*, Lake Charles, LA
09/2008 – 08/2009 **Software Engineer**, *FPT Software Company*, Ho Chi Minh City, Vietnam

IBM RESEARCH ACCOMPLISHMENTS

2024 Granite Time Series Foundation Models (O-level)
2023 Research Contributions to Time Series Foundation Models (A-level)
2022 Federated Learning Security and Privacy (O-level)
2022 Dynamic Approaches for Machine Learning (A-level)
2022 Regression Optimization for Heavy Processing Industries (A-level)
2022 Combinatorial Sparsity for AI (A-level)
2021 Stochastic Gradient Methods: Theory and Applications (A-level)
2019 SROM: Smarter Resource & Operations Management (A-level)

HONORS & AWARDS

2024 INFORMS Senior Member
2023 2022 Pat Goldberg Memorial Best Paper Award
2023 IBM 9th Plateau Invention Achievement Award
2023 IBM Outstanding Technical Achievement Award, “*Dynamic Approaches for Machine Learning*”
2023 IBM Outstanding Technical Achievement Award, “*Regression Optimization for Heavy Processing Industries*”
2023 IBM Outstanding Technical Achievement Award, “*Federated Learning Security and Privacy*”
2023 IBM Outstanding Technical Achievement Award, “*Combinatorial Sparsity for AI*”
2023 IBM 8th Plateau Invention Achievement Award
2022 Winner of U.S. PETs Prize Challenge (Concept Papers)
2022 IBM Master Inventor
2022 IBM 7th Plateau Invention Achievement Award
2022 IBM 6th Plateau Invention Achievement Award
2022 IBM Outstanding Technical Achievement Award, “*Stochastic Gradient Methods: Theory and Applications*”
2022 IBM 5th Plateau Invention Achievement Award
2022 IBM 4th Plateau Invention Achievement Award
2021 IBM 3rd Plateau Invention Achievement Award
2020 IBM 2nd Plateau Invention Achievement Award
2020 IBM Research Division Award
2020 IBM Outstanding Technical Achievement Award, “*SROM: Smarter Resource & Operations Management*”
2020 IBM 1st Plateau Invention Achievement Award
2019 NeurIPS 2019 Top Reviewers
2019 Elizabeth V. Stout Dissertation Award, *Lehigh University*, Bethlehem, PA
2018 Van Hoesen Family Best Publication Award, *Lehigh University*, Bethlehem, PA
2016 – 2017 Dean’s Doctoral Fellowship (RCEAS), *Lehigh University*, Bethlehem, PA
2014 – 2015 Dean’s Doctoral Assistantship, *Lehigh University*, Bethlehem, PA
2014 Beta Gamma Sigma (Academic Honor Society)